

Lab Practical– 3

Note: Do not use any machine learning library. Implement the following functions from scratch.

Question 1:

Implement the AND function using a single sigmoid neuron.

1. Model the neuron output as:

$$\hat{y} = \sigma(w_1x_1 + w_2x_2 + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}}$$

2. Use the squared error loss and update w_1, w_2, b using gradient descent.
3. Train the neuron on all four input-output pairs of OR:

$$(0, 0, 0), \quad (0, 1, 0), \quad (1, 0, 0), \quad (1, 1, 1)$$

4. Report the learned weights, bias, and predicted outputs.

Question 2:

Implement the following Boolean function using a single sigmoid neuron.

$$y = x_1 \text{ OR } x_2 \text{ OR } x_3,$$

where $x_1, x_2, x_3 \in \{0, 1\}$.

1. Model the neuron output as

$$\hat{y} = \sigma(w_1x_1 + w_2x_2 + w_3x_3 + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}}.$$

2. Use the squared error loss and implement gradient descent to update the weights w_1, w_2, w_3 and bias b .
3. Train the neuron on all 8 possible input-output combinations of x_1, x_2, x_3 .
4. After training, compute the predicted outputs and report the learned weights, bias, and accuracy.

Question 3:

Implement the XOR function using a feedforward neural network with sigmoid activation neurons.

1. Design a neural network with:
 - 2 input neurons (for the two binary inputs x_1 and x_2)
 - 1 hidden layer with an appropriate number of neurons
 - 1 output neuron (for the XOR output)

2. Use the sigmoid function as the activation function for all neurons:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

3. Define the squared error loss function:

$$L = \frac{1}{2} \sum_{i=1}^4 (\hat{y}_i - y_i)^2$$

where (x_1, x_2, y) are the four possible input-output pairs of XOR:

$$(0, 0, 0), \quad (0, 1, 1), \quad (1, 0, 1), \quad (1, 1, 0)$$

4. Implement forward propagation and backpropagation to update the weights and biases using gradient descent.
5. Train the network until the loss converges.
6. After training, verify the network by computing the output for all four input combinations and report the predicted outputs.